[INCH-POUND]
A-A-60017
12 August 2020

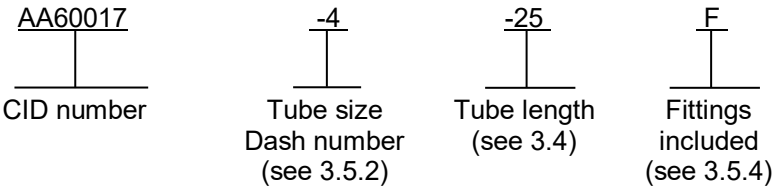
COMMERCIAL ITEM DESCRIPTION

TUBE AND FITTINGS, METALLIC, BRAKE, FUEL, AND TRANSMISSION LINES, NICKEL-COPPER ALLOY, GENERAL REQUIREMENTS FOR

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers the general requirements for tube and fittings, metallic, brake, fuel, and transmission lines, nickel-copper alloy. Tube and fittings, metallic, brake, fuel, and transmission lines, nickel-copper alloy covered by this CID are intended for commercial/industrial applications.

2. CLASSIFICATION/PART OR IDENTIFICATION NUMBER (PIN). This CID uses a classification system which is included in the PIN as shown in the following example (see 7.1).



PIN examples:

AA60017-4-25F - Specifies tubing ¼ inch OD twenty five feet long, with 8 inverted flare nuts.

AA60017-4-25 - Specifies tubing ¼ inch OD twenty five feet long, with no inverted flare nuts.

3. SALIENT CHARACTERISTICS.

3.1 Interface and physical dimensions. Tube and fittings, metallic, brake, fuel, and transmission lines, nickel-copper alloy supplied to this CID shall be as specified herein. A flaring tool is required assemble the tube assembly.

3.2 CID specification sheet. The family of tube and fittings, metallic, brake, fuel, and transmission lines, nickel-copper alloy shall be in accordance with the requirements specified herein.

3.3 Design and construction. Tube and fittings, metallic, brake, fuel, and transmission lines, nickel-copper alloy supplied to this CID are for use with brake, fuel, and transmission lines, they are designed to be used where the underbody of the vehicle is subjected to the harshest environments, where the longest life-span for lines is required, and/or where direct OEM replacement is desired.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data that may improve this document should be sent to: DLA Land and Maritime, ATTN: VAI, P.O. Box 3990, Columbus OH 43218-3990, or fluidflow@dla.mil. Since contact information can change you may want to verify the currency of the address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A

FSC 4710

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3.4 Tube. Tubing shall be supplied in coils 10, 15, 25, 50, or 100 feet as specified in the PIN. Tubing shall be seamless - extruded from an ingot, not rolled. Length tolerance shall be in accordance with ASTM B251/B251M. Note: reference tables in ASTM B251/B251M for ASTM B466/466M type tubing.

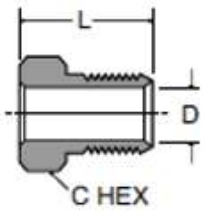
3.4.1 Tube materials. Tube materials shall be 90-10 nickel-copper alloy brake tubing in accordance with SAE J1650 or ASTM B466/B466M alloy UNS70600.

3.5 Inverted flare nut. Fittings supplied to this CID are “Long Nut” inverted flare.

3.5.1 Inverted flare nut material. Fitting materials shall be brass in accordance with SAE J512.

3.5.2 Tube size. Tube size dash number is the size of tube expressed in sixteenths of an inch, tubes are measured on outside diameter, see figure 1 for tube size dash number.

3.5.3 Inverted flare nut dimensions. Fitting dimensions shall be in accordance with SAE J512 040110, see figure 1.



Tube size dash number	SAE J512 Fitting/tube size O.D.	Straight thread	C Hex	L	D
-3	3/16	3/8-24	3/8	.53	.188
-4	1/4	7/16-24	7/16	.54	.188
-5	5/16	1/2-20	1/2	.59	.250
-6	3/8	5/8-18	5/8	.66	.312
-7	7/16	11/16-18	11/16	.75	.375
-8	1/2	3/4-18	3/4	.81	.576

NOTE: Dimensions and tolerances not show shall be in accordance with SAE J512 040110.

Figure 1. Inverted flare nuts.

3.5.4 Inverted flare nuts supplied. When specified in the PIN, by the “F” designator, the following number of inverted flare nuts shall be furnished with the coil tubing. The inverted flare nuts shall fit the tubing size specified in the PIN, see table I..

Table I. Tubing coil lengths and number of brass flare nuts.

Tube length PIN Code	Supplied parts
-10F	10 foot coil comes with 4 inverted flare nuts.
-15F	15 foot coil comes with 6 inverted flare nuts.
-25F	25 foot coil comes with 8 inverted flare nuts.
-50F	50 foot coil comes with 16 inverted flare nuts.
-100F	100 foot coil comes with 32 inverted flare nuts.

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3.6 Finish. Finish shall be in accordance with SAE J512.

3.7 Usage. 90-10 copper-nickel corrosion resistant tube can be used for brake, fuel, and transmission lines. DOT approved for hydraulic brake systems and bends easier than steel tubing.

3.7.1 Working pressures. Working pressure see table II.

TABLE II. Working Pressure.

Tube O.D	PSI	Tube wall
3/16	3200	.028
1/4	2300	.028
5/16	1800	.028
3/8	1450	.028
1/2	1050	.028
5/8	1450	.048

3.8 Cleaning. The inside and outside surfaces of the finished tubing shall be cleaned in accordance with SAE J1650. Tubing and fittings shall be clean, and free from grease, oxide scale, carbon deposits, and any other contamination that cannot be readily removed by cleaning agents normally used in manufacturing plants.

3.9 Marking. Brake, fuel, and transmission line kits supplied to this CID shall be marked with the manufacturer's (MFR's) standard commercial PIN. (NOTE: The part number marked on the unit pack shall be the CID PIN.)

3.10 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.11 Workmanship. Brake, fuel, and transmission line kits shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with 23.403 of the Federal Acquisition Regulation (FAR).

5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer’s own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

5.2 Certification. Certification must be done with the procuring activity approval. The contractor shall certify that the product offered meets the salient characteristics of the description and conforms to the producer’s own drawings, specifications, standards, and quality assurance practices, and is the same as the product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

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6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES.

7.1 PIN. The PIN should be used for Government purposes to buy commercial products to this CID. See section 2 for PIN format examples.

7.2 Commercial and Government Entity (CAGE) code. For ordering purposes, inventory control, and submission of these tube and fittings, metallic, brake, fuel, and transmission lines, nickel-copper alloy to DLA Land and Maritime under the Parts Management Advisory Team (PMAT) evaluation program, CAGE code 58536 should be used.

7.3 Source of documents.

FEDERAL REGULATIONS

FAR – Federal Acquisition Regulations (FAR)

(Copies of these documents are available online at https://www.acquisition.gov/comp/far/index.html.)

ASTM INTERNATIONAL

ASTM B251/B251M - Standard Specification for General Requirements for Wrought Seamless Copper and Copper-Alloy Tube
ASTM B466/B466M - Standard Specification for Seamless Copper-Nickel Pipe and Tube

(Copies of these documents are available online at https://www.astm.org.)

SAE INTERNATIONAL

SAE J512 - Automotive Tube Fittings
SAE J1650 - Seamless Copper-Nickel Tubing

(Copies of these documents are available online at https://www.sae.org.)

7.4 Ordering data. The contract or order should specify the following:

a. CID document number, revision, and CID PIN

b. Product conformance provisions.

c. Packaging requirements.

d. Test certificates of the tubing, if required, see SAE J1650.

7.5 Government users. To acquire information on obtaining these tube and fittings, metallic, brake, fuel, and transmission lines, nickel-copper alloy from the Government inventory system, contact DLA Land and Maritime, ATTN: DLA Land and Maritime Call Center (-NAB), PO BOX 3990, Columbus, OH 43218-3990, or telephone (614) 692-2271 or (614) 692-3191.

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7.6 Commercial products. As part of the market analysis and research effort, this CID was coordinated with the following manufacturers of commercial products. At the time of CID preparation and coordination, these manufacturers were known to have commercial products that would meet the requirements of this CID. (NOTE: This information should not be considered as a list of approved manufacturers or be used to restrict acquisition to only the manufacturers shown.)

MFR's CAGE	MFR's name and address
4GB20	AGS Company P.O. Box 729 Muskegon, MI 49433 Phone 1-800-253-0403 E-mail: customerservice@agscompany.com https://agscompany.com
56L80	Federal Hill Trading Company 195 Federal Hill Road Oxford, MA 01540-1303 Phone 508-987-2660 FAX 508-987-2661 E-mail: fedhillusa@gmail.com https://store.fedhillusa.com/
1EJL1	NAPA 2105 HWY 180 E Silver City, MN 88061-7787 Phone 1-800-538-6272 E-mail: customersupport@napaonline.com https://www.napaonline.com/

7.7 Government users. To acquire information on obtaining these tube and fittings, metallic, brake, fuel, and transmission lines, nickel-copper alloy from the Government inventory system, contact DLA Land and Maritime, ATTN Land Weapons System Division (-QL), P.O. Box 3990, Columbus, OH 43218-3990 or E-mail - DSCC-QLR.Chief@dla.mil, telephone (614) 692-4702 or DSN 850-3796.

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TUBE AND FITTINGS, METALLIC, BRAKE, FUEL, AND TRANSMISSION LINES, NICKEL-COPPER ALLOY, INSTALLATION PROCEDURE

A.1 SCOPE.

A.1.1 Scope. This commercial item description (CID) covers the general requirements for tube and fittings, metallic, brake, fuel, and transmission lines, nickel-copper alloy, installation procedure covered by this CID are intended for commercial/industrial applications. This appendix is not a mandatory part of the commercial item description (CID).

A.1.2 Intended use. The long nut specified in the CID offers additional vibration capacity.

A1.3 Assembly instructions:

a. Cut tubing squarely and clean tube end thoroughly to remove burrs.
b. Place nut onto tube. Place threaded end of nut toward end of tube.
c. Flare tube end with flaring tool to provide a 37º or 45° flare, as required by the application.
d. Lubricate threads and assemble to fitting body.
e. Nut should be turned hand tight.
f. Tighten with a wrench until a solid feeling is encountered. From that point, apply a one-six turn.
g. Flare fittings are easy to disassemble and may be reassembled repeatedly, for a leak-proof connection.

NOTE: Do not over-torque as it may damage the fitting or split the tubing.

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MILITARY INTERESTS:

Custodians:
Army - AT
Navy - AS
Air Force - 71
DLA - CC

Review activities:
Army - AV, MI
Navy - MC, SH
Air Force - 11

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FAS

Preparing Activity:

DLA - CC

Project 4710-2020-002

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